



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

**Constituent of Symbiosis International (Deemed University),
Pune**

Subject: Computer Vision Laboratory

Experiment - 10

Aim: To use a pre-trained DenseNet model for feature extraction and classification. **Software**

Tools Required: Google Colab

Theory:

1. Introduction to Transfer Learning

Transfer learning is a deep learning technique where a model developed for one task is reused as the starting point for another related task. Instead of training a convolutional neural network (CNN) from scratch—which requires large datasets and extensive computational resources—we use a model that has already been pre-trained on a large dataset such as ImageNet. This pre-trained model has already learned to detect low-level features (edges, textures, colors) and high-level features (shapes, objects) that can generalize well to new problems.

2. Overview of DenseNet Architecture

DenseNet (Densely Connected Convolutional Network) is a deep CNN architecture proposed to improve information and gradient flow between layers. In DenseNet:

- Each layer receives feature maps from all previous layers as inputs.
- The outputs of each layer are passed to all subsequent layers.

This dense connectivity pattern:

- Encourages feature reuse, leading to efficient learning.
- Reduces the number of parameters compared to traditional CNNs.
- Improves gradient propagation, avoiding vanishing gradient issues.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

3. Why Use DenseNet for Transfer Learning

DenseNet models (e.g., DenseNet121, DenseNet169, DenseNet201) are pre-trained on ImageNet, a massive dataset with over a million labeled images across 1,000 classes.

Constituent of Symbiosis International (Deemed University), Pune

These pre-trained weights allow the network to serve as a powerful generic feature extractor for various image classification tasks, even when the new dataset is small.

Advantages:

- Reduced training time.
- Better performance with limited data.
- Feature generalization across different domains.

4. Feature Extraction using DenseNet

In feature extraction, the top (fully connected) classification layer of DenseNet is removed because it is specific to the original ImageNet classes (e.g., dog, car, etc.). We retain the convolutional base, which outputs rich and high-dimensional feature maps representing the input images.

The procedure is as follows:

1. Load the pre-trained DenseNet model without its top layer (`include_top=False`).
2. Freeze the convolutional layers so their weights are not updated during training.
3. Pass the new dataset's images through the DenseNet base to extract feature vectors.
4. Feed these features into a custom classifier such as:
 - A simple fully connected neural network (MLP).
 - A Support Vector Machine (SVM).



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

- A Logistic Regression classifier.

This process leverages the general visual representations learned by DenseNet while customizing only the final classification stage for the new dataset.

5. Classification Layer (Custom Classifier)

After feature extraction, the classifier learns to map the extracted features to the specific classes of the target dataset.

Constituent of Symbiosis International (Deemed University), Pune

If using an MLP:

- The extracted features are input to a Dense layer with activation functions like ReLU and Softmax.
- The model is trained on the target dataset labels.

If using an SVM:

- The feature vectors are used to train a linear or kernel-based SVM for classification.

6. Model Evaluation

Once the classifier is trained, its performance is evaluated using:

- Accuracy vs Epochs graph: to visualize learning progression.
- Confusion matrix: to analyze misclassifications and class-wise accuracy.

7. Advantages of Using DenseNet for Feature Extraction

- Efficient feature reuse improves representational power.
- Requires fewer parameters than other deep CNNs.
- Performs well even on small datasets.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

- Faster convergence and less overfitting due to dense connectivity.

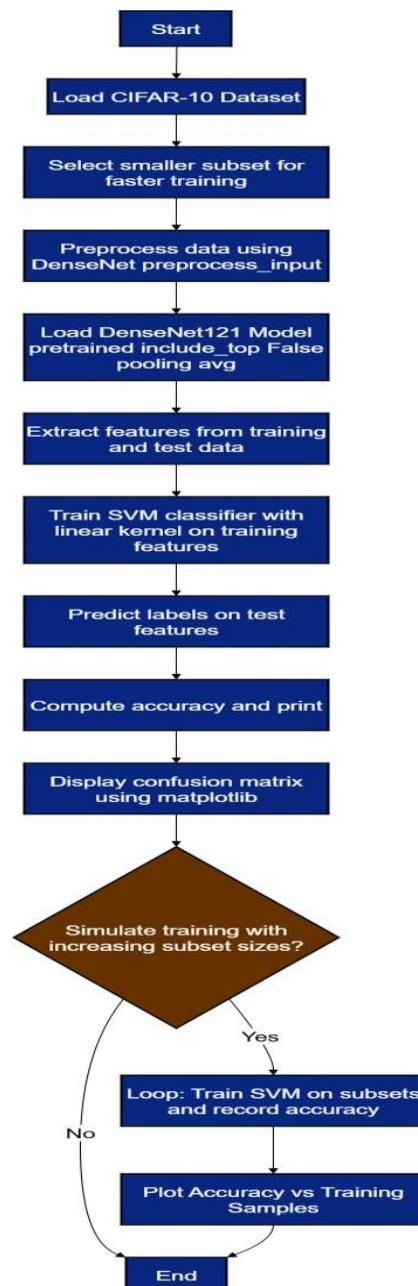
**Constituent of Symbiosis International (Deemed University),
Pune**

Flowchart



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Part-1



**Constituent of Symbiosis International (Deemed University),
Pune**



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Program with Output:

Program:

```
from keras.applications import DenseNet121
model = DenseNet121(weights='imagenet', include_top=False, pooling='avg', input_shape=(32, 32, 3))

from keras.datasets import cifar10
from keras.applications.densenet import preprocess_input
import numpy as np
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

(X_train, y_train), (X_test, y_test) = cifar10.load_data()

# Smaller subset for faster training
X_train, y_train = X_train[:200], y_train[:200]
X_test, y_test = X_test[:50], y_test[:50]

# Preprocess to match ImageNet input scaling
X_train = preprocess_input(X_train.astype('float32'))
X_test = preprocess_input(X_test.astype('float32'))

features_train = model.predict(X_train, verbose=1)
features_test = model.predict(X_test, verbose=1)

clf = SVC(kernel='linear')
clf.fit(features_train, y_train.ravel())

y_pred = clf.predict(features_test)
acc = accuracy_score(y_test, y_pred)
print("Accuracy:", acc)

# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap='Blues')
plt.title("Confusion Matrix")
plt.show()

accuracies = []
for i in range(20, 201, 20):
    clf = SVC(kernel='linear')
    clf.fit(features_train[:i], y_train[:i].ravel())
    y_pred = clf.predict(features_test)
    accuracies.append(accuracy_score(y_test, y_pred))

plt.plot(range(20, 201, 20), accuracies, marker='o')
plt.xlabel("Training samples (simulated epochs)")
plt.ylabel("Accuracy")
plt.title("Accuracy vs Training Size (simulated Epochs)")
plt.grid(True)
plt.show()
```

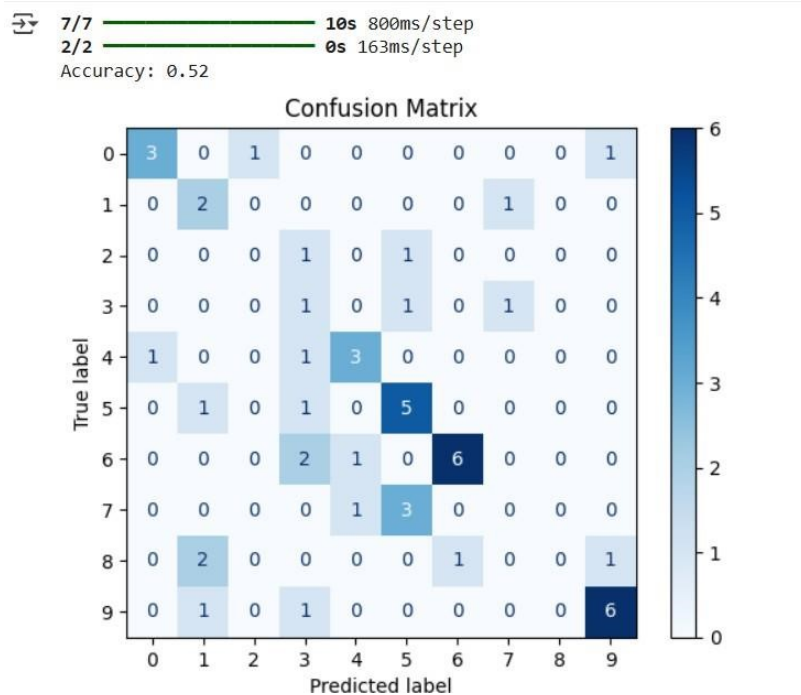


Name-Janay rawal
PRN-22070123049
ENTC-A (G1)

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University),
Pune

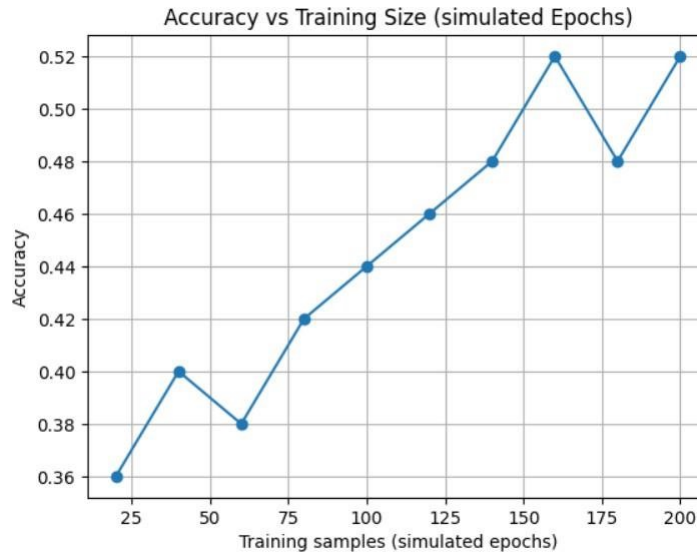
Output:





Name-Janay rawal
PRN-22070123049
ENTC-A (G1)

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE



**Constituent of Symbiosis International (Deemed University),
Pune**

Conclusion/Inferences:

Using a pre-trained DenseNet model for feature extraction and classification effectively leverages the power of transfer learning. By utilizing DenseNet's pre-learned feature representations, the model can achieve high accuracy with reduced training time and computational effort. This approach allows efficient adaptation to new datasets while maintaining strong generalization, making it an excellent choice for image classification tasks with limited data.